

条件极值:

设 $\underbrace{f(x, y, u, v)}_{\text{目标函数}}, \underbrace{g(x, y, u, v), h(x, y, u, v)}_{\text{约束函数}}$ 是连续可微.

求 $f(x, y, u, v)$ 在约束条件

$$\begin{cases} g(x, y, u, v) = 0, \\ h(x, y, u, v) = 0 \end{cases} \Rightarrow \exists \begin{cases} u = u(x, y) \\ v = v(x, y) \end{cases}$$

下求极值. 假设 $J = \frac{\partial(g, h)}{\partial(u, v)} \neq 0$. 由隐函数定理.

$\exists$ 隐函数 $\begin{cases} u = u(x, y), \\ v = v(x, y), \end{cases}$ 连续可微. 问题转化为求

$f(x, y, u(x, y), v(x, y))$ 的无条件极值. 若在 (x_0, y_0) 取极值, 则

$$\begin{cases} u_0 = u(x_0, y_0) \\ v_0 = v(x_0, y_0) \end{cases} \Rightarrow \text{则 } \underline{P_0(x_0, y_0, u_0, v_0)}: \begin{cases} g(x_0, y_0, u_0, v_0) = 0 \\ h(x_0, y_0, u_0, v_0) = 0 \end{cases}$$

$$\text{即 } g(P_0) = h(P_0) = 0$$

$f(x_0, y_0, u_0, v_0)$ 是极值. 则

$F(x, y) \triangleq f(x, y, u(x, y), v(x, y))$ 在 (x_0, y_0) 取极值.

$$\Rightarrow \begin{cases} F_x(x_0, y_0) = 0, \\ F_y(x_0, y_0) = 0 \end{cases}$$

$$F_x(x_0, y_0) = f_x(P_0) + f_u(P_0) \cdot u_x + f_v(P_0) \cdot v_x = 0$$

$$F_y(x_0, y_0) = f_y(P_0) + f_u(P_0) \cdot u_y + f_v(P_0) \cdot v_y = 0$$

$$dF(x_0, y_0) = F_x(x_0, y_0) dx + F_y(x_0, y_0) dy = 0.$$

$$= (f_x(p_0) + f_u(p_0) u_x + f_v(p_0) v_x) dx + (f_y(p_0) + f_u(p_0) u_y + f_v(p_0) v_y) dy = 0$$

$$= f_x(p_0) dx + f_u(p_0) u_x dx + f_v(p_0) v_x dx + f_y(p_0) dy + f_u(p_0) u_y dy + f_v(p_0) v_y dy$$

$$= f_x(p_0) dx + f_y(p_0) dy + f_u(p_0) du + f_v(p_0) dv$$

$$du = u_x dx + u_y dy$$

$$dv = v_x dx + v_y dy$$

$$dF|_{p_0} = (f_x dx + f_y dy + f_u du + f_v dv)|_{p_0} = 0. \quad (*)_1$$

$$(F(x, y) : \text{and } (x_0, y_0) : dF|_{(x_0, y_0)} = (F_x dx + F_y dy)|_{(x_0, y_0)} = 0)$$

$$\text{例 14: } g(x, y, u(x, y), v(x, y)) = 0$$

$$h(x, y, u(x, y), v(x, y)) = 0$$

$$0 = dg = g_x dx + g_y dy + g_u du + g_v dv = 0 \quad (*)_2$$

$$(du = u_x dx + u_y dy, dv = v_x dx + v_y dy).$$

$$0 = dh = h_x dx + h_y dy + h_u du + h_v dv = 0 \quad (*)_3$$

$$(*)_1 + \lambda_1 (*)_2 + \lambda_2 (*)_3 \Rightarrow$$

$$d(F + \lambda_1 g + \lambda_2 h) \Big|_{p_0} = (f_x + \lambda_1 g_x + \lambda_2 h_x) \Big|_{p_0} dx \\ + (f_y + \lambda_1 g_y + \lambda_2 h_y) \Big|_{p_0} dy \\ + (f_u + \lambda_1 g_u + \lambda_2 h_u) \Big|_{p_0} du \\ + (f_v + \lambda_1 g_v + \lambda_2 h_v) \Big|_{p_0} dv = 0$$

找 λ_1, λ_2 使得

$$\begin{cases} (f_u + \lambda_1 g_u + \lambda_2 h_u) \Big|_{p_0} = 0 \\ (f_v + \lambda_1 g_v + \lambda_2 h_v) \Big|_{p_0} = 0 \end{cases} \quad \nexists \lambda_1, \lambda_2 \quad (*)_{\neq}$$

使之为零

$$\Rightarrow d(F + \lambda_1 g + \lambda_2 h) \Big|_{p_0} = (f_x + \lambda_1 g_x + \lambda_2 h_x) \Big|_{p_0} dx \\ + (f_y + \lambda_1 g_y + \lambda_2 h_y) \Big|_{p_0} dy = 0$$

但: dx, dy : 任意独立变量:

$$\Rightarrow \begin{cases} (f_x + \lambda_1 g_x + \lambda_2 h_x) \Big|_{p_0} = 0 \\ (f_y + \lambda_1 g_y + \lambda_2 h_y) \Big|_{p_0} = 0 \end{cases}$$

$$(*)_{\neq} \text{ 改为: } \begin{cases} g_u(p_0)\lambda_1 + h_u(p_0)\lambda_2 = -f_u(p_0) \\ g_v(p_0)\lambda_1 + h_v(p_0)\lambda_2 = -f_v(p_0) \end{cases}$$

这是关于 λ_1, λ_2 的线性方程组: 系数行列式

$$\begin{vmatrix} g_u & h_u \\ g_v & h_v \end{vmatrix} \Big|_{P_0} \stackrel{\text{行列变换}}{=} \begin{vmatrix} g_u & g_v \\ h_u & h_v \end{vmatrix} \Big|_{P_0} \stackrel{\text{已知条件}}{\downarrow} \frac{\partial(g, h)}{\partial(u, v)} \Big|_{P_0} \neq 0$$

$$\Rightarrow \exists \lambda_1, \lambda_2.$$

$$\text{在 } (x_0, y_0, u_0, v_0, \lambda_1, \lambda_2): \begin{cases} (f_x + \lambda_1 g_x + \lambda_2 h_x) \Big|_{P_0} = 0 \\ (f_y + \lambda_1 g_y + \lambda_2 h_y) \Big|_{P_0} = 0 \\ (f_u + \lambda_1 g_u + \lambda_2 h_u) \Big|_{P_0} = 0 \\ (f_v + \lambda_1 g_v + \lambda_2 h_v) \Big|_{P_0} = 0 \end{cases}$$

于是求 $f(x, y, u, v)$ 在约束条件 $\begin{cases} g=0 \\ h=0 \end{cases}$ 下的极值所转化为求

$$L = L(x, y, u, v, \lambda_1, \lambda_2)$$

$$\triangleq f(x, y, u, v) + \lambda_1 g(x, y, u, v) + \lambda_2 h(x, y, u, v)$$

$\therefore$ 临界点. 求 L 之临界点:

$$(*) \begin{cases} \left(\frac{\partial L}{\partial x} = f_x + \lambda_1 g_x + \lambda_2 h_x \right) \Big|_{P_0} = 0 \\ \left(\frac{\partial L}{\partial y} = f_y + \lambda_1 g_y + \lambda_2 h_y \right) \Big|_{P_0} = 0 \\ \left(\frac{\partial L}{\partial u} = f_u + \lambda_1 g_u + \lambda_2 h_u \right) \Big|_{P_0} = 0 \\ \left(\frac{\partial L}{\partial v} = f_v + \lambda_1 g_v + \lambda_2 h_v \right) \Big|_{P_0} = 0 \end{cases}$$

$$\left\{ \begin{array}{l} \frac{\partial L}{\partial \lambda_1} = g|_{p_0} = 0 \\ \frac{\partial L}{\partial \lambda_2} = h|_{p_0} = 0 \end{array} \right\} \leftarrow \text{此即约束条件.}$$

于是有

定理: 设 $f(x, y, u, v)$, $g(x, y, u, v)$, $h(x, y, u, v)$ 连续

可微, Jacobi 行列式 $\frac{\partial(g, h)}{\partial(u, v)} \neq 0$.

如果在约束条件 $\begin{cases} g(x, y, u, v) = 0 \\ h(x, y, u, v) = 0 \end{cases}$ 下 $f(x, y, u, v)$

在点 $p_0(x_0, y_0, u_0, v_0)$ 取得极值, 则 $\exists \lambda_1, \lambda_2 \in \mathbb{R}$ 使

得 $(p_0, \lambda_1, \lambda_2) = (x_0, y_0, u_0, v_0, \lambda_1, \lambda_2) \triangleq$ Lagrange 点

$L(x, y, u, v, \lambda_1, \lambda_2) = f + \lambda_1 g + \lambda_2 h$ 的临界点, 即

在该点处 $\frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} = \frac{\partial L}{\partial u} = \frac{\partial L}{\partial v} = g = h = 0$.

这样的求条件极值的方法称为 Lagrange 乘子法, $L(x, y, u, v, \lambda_1, \lambda_2)$ 称为 Lagrange 函数, λ_1, λ_2 称为 Lagrange 乘子.

更一般地: $F(x), x \in D, D \subset \mathbb{R}^n$ 且 $1 \leq m < n$.

求 $F(x)$ 在约束条件 $\begin{cases} g_1(x) = 0 \\ g_2(x) = 0 \\ \vdots \\ g_m(x) = 0 \end{cases}$ 下之极值:

$$\text{令 } L(x, \lambda) = F(x) + \sum_{j=1}^m \lambda_j g_j(x), x = (x_1, x_2, \dots, x_n) \in D \subset \mathbb{R}^n$$

$\lambda = (\lambda_1, \lambda_2, \dots, \lambda_m)$. $L(x, \lambda)$ 称为 Lagrange 函数.

假设 $g_j(x)$ ($1 \leq j \leq m$): 满足. Jacobi 矩阵非退化:

$$\begin{pmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \dots & \frac{\partial g_1}{\partial x_n} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \dots & \frac{\partial g_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_m}{\partial x_1} & \frac{\partial g_m}{\partial x_2} & \dots & \frac{\partial g_m}{\partial x_n} \end{pmatrix}_{m \times n} \triangleq \left(\frac{\partial g_j}{\partial x_i} \right)_{m \times n}$$

$\updownarrow$
 满秩: $m < n$.
 秩为 m :

不妨设前 m 行 m 列之行列式 $\neq 0$, 即

$$\frac{\partial (g_1, g_2, \dots, g_m)}{\partial (x_1, x_2, \dots, x_m)} = \begin{vmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \dots & \frac{\partial g_1}{\partial x_m} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \dots & \frac{\partial g_2}{\partial x_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_m}{\partial x_1} & \frac{\partial g_m}{\partial x_2} & \dots & \frac{\partial g_m}{\partial x_m} \end{vmatrix} \neq 0.$$

由隐函数定理: $u > (x_{m+1}, x_{m+2}, \dots, x_n) = z$ 为独立变量,

$$\Rightarrow \left\{ \begin{array}{l} x_1 = x_1(z) \\ x_2 = x_2(z) \\ \vdots \\ x_m = x_m(z) \end{array} \right.$$

$g_j(x)$ 连续可微

$\Rightarrow x_j(z)$ 连续可微

$\rightarrow n-m$ 个自变量

$$F(z) = f(x) = f(x_1(z), x_2(z), \dots, x_m(z), \underbrace{x_{m+1}, \dots, x_n}_{n-m \text{ 个自变量}})$$

$z = (x_{m+1}, x_{m+2}, \dots, x_n)$ 为独立变量

$F(z)$ 是 $n-m$ 元函数, 在极值点处

$$dF = 0 = df = \sum_{j=1}^n \frac{\partial f}{\partial x_j} dx_j = 0.$$

类似于前面关于 $f(x, y, u, v)$ 在约束条件 $\begin{cases} g(x, y, u, v) = 0 \\ h(x, y, u, v) = 0 \end{cases} \rightarrow$

极值问题的讨论, 求 $f(x)$ 在 $g_j(x) = 0 (j=1, \dots, m)$.

之极值问题. 就化为求 $L = f(x) + \sum_{j=1}^m \lambda_j g_j(x)$ 极值问题.

求 $L(x, \lambda)$ 之临界点:

$$\begin{cases} \frac{\partial L}{\partial x_i} = \frac{\partial f}{\partial x_i} + \sum_{j=1}^m \lambda_j \frac{\partial g_j}{\partial x_i} = 0, & i=1, 2, \dots, n, \\ g_j(x) = 0 & j=1, 2, \dots, m. \end{cases}$$

由此求出驻点 $(x_0 = (x_1^0, x_2^0, \dots, x_n^0); \lambda_0 = (\lambda_1^0, \lambda_2^0, \dots, \lambda_m^0))$.

驻点是否极值点? 极值的充分条件:

再回顾极值问: 设 $f(x, y)$ 二阶连续可微,
在 $P_0(x_0, y_0)$ 取极值. 则 $f_x(x_0, y_0) = 0, f_y(x_0, y_0) = 0$,

i.e. $\nabla f(x_0, y_0) = \vec{0}$, 对 $x = x_0 + dx, y = y_0 + dy$,

$\rho = \sqrt{dx^2 + dy^2}$, 由 Taylor 公式,

$$f(x, y) = f(x_0, y_0) + \nabla f(x_0, y_0) \cdot (\Delta x, \Delta y) + \frac{1}{2!} (\Delta x, \Delta y) H \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} + o(\rho^2)$$

$$H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{pmatrix} \Big|_{(x_0, y_0)} \quad \text{—— Hesse 矩阵}$$

$$\Delta f = f(x, y) - f(x_0, y_0) = \frac{1}{2} (dx, dy) \cdot H \begin{pmatrix} dx \\ dy \end{pmatrix} + o(\rho^2)$$

$= \frac{1}{2} d^2 f$ —— 用二阶微分来判别

$$\Rightarrow d^2 f = (dx, dy) \begin{pmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix}$$

$$= f_{xx} dx^2 + 2 f_{xy} dx dy + f_{yy} dy^2$$

$$\Rightarrow \begin{cases} > 0 \Leftrightarrow H \text{ 正定: } f(x_0, y_0) \text{ 是极小值.} \\ < 0 \Leftrightarrow H \text{ 负定: } f(x_0, y_0) \text{ 是极大值.} \end{cases}$$

设在约束条件 $\begin{cases} g(x, y, u, v) = 0, \\ h(x, y, u, v) = 0 \end{cases}$ 下, $f(x, y, u, v)$ 在

$P_0(x_0, y_0, u_0, v_0)$ 取极值.

在此约束条件下, $\exists$ (隐函数) $u = u(x, y), v = v(x, y)$

则 Lagrange 函数

$$L = L(x, y, u, v, \lambda_1, \lambda_2) = f + \lambda_1 g + \lambda_2 h$$

化为 $L(x, y, u, v, \lambda_1, \lambda_2) = f(x, y, u(x, y), v(x, y))$

$$F(x, y) \triangleq L(x, y, u(x, y), v(x, y), \lambda_1, \lambda_2) = f(x, y, u(x, y), v(x, y))$$

在 (x_0, y_0) 取得极值 $\Rightarrow$

$$dF|_{(x_0, y_0)} = F_x(x_0, y_0) dx + F_y(x_0, y_0) dy = 0$$

这是必要条件. 再计算 $F(x, y)$ 的全微分:

$$\therefore dF(x, y) = \frac{\partial L}{\partial x} dx + \frac{\partial L}{\partial y} dy + \frac{\partial L}{\partial u} du + \frac{\partial L}{\partial v} dv$$

其中 dx, dy 是独立变量 $\rightarrow$ 独立变量

$$du = u_x dx + u_y dy, \quad dv = v_x dx + v_y dy$$

是 (x, y, dx, dy) 的函数

二阶微分:

$$d^2F(x,y) = d\left(\frac{\partial L}{\partial x}\right)dx + d\left(\frac{\partial L}{\partial y}\right)dy + d\left(\frac{\partial L}{\partial u}\right)du \\ + \frac{\partial L}{\partial u} d(du) + d\left(\frac{\partial L}{\partial v}\right)dv + \frac{\partial L}{\partial v} d(dv)$$

在极值点 (x_0, y_0) 处,

在 P_0 处为 0.

$$d^2F(x_0, y_0) = d(L_x)|_{P_0}dx + d(L_y)|_{P_0}dy + d(L_u)|_{P_0}du \\ + \frac{\partial L}{\partial u}|_{P_0}d^2u + d(L_v)|_{P_0}dv + \frac{\partial L}{\partial v}|_{P_0}d^2v$$

由必要条件, 在极值点处, 前者为零

$$\therefore d^2F = (dL_x)|_{P_0}dx + d(L_y)|_{P_0}dy + d(L_u)|_{P_0}du + d(L_v)|_{P_0}dv \\ = d^2L.$$

$\Rightarrow$ 条件极值的充分条件

$$\begin{cases} d^2L > 0, \text{ 正定: } f(P_0) \text{ 是极小值,} \\ d^2L < 0, \text{ 负定: } f(P_0) \text{ 是极大值.} \end{cases}$$

例 求 $f(x,y,z,t) = x+y+z+t$ 在约束条件 $xyzt = c$ ($c > 0$) 下 $x, y, z, t > 0$ 时的极值.

例: 设 $L = x + y + z + t + \lambda(xyzt - c^4)$, 求 L 的极值:

$$\begin{cases} \frac{\partial L}{\partial x} = 1 + \lambda yzt = 0 \leftarrow \cdot x : x + \lambda xyzt = 0 \\ \frac{\partial L}{\partial y} = 1 + \lambda xzt = 0 \leftarrow \cdot y : y + \lambda xyzt = 0 \\ \frac{\partial L}{\partial z} = 1 + \lambda xyt = 0 \leftarrow \cdot z : z + \lambda xyzt = 0 \\ \frac{\partial L}{\partial t} = 1 + \lambda xyz = 0 \leftarrow \cdot t : t + \lambda xyzt = 0 \\ \frac{\partial L}{\partial \lambda} = xyzt - c^4 = 0 \end{cases}$$

$$\Rightarrow x = y = z = t = -\lambda xyzt \Rightarrow x^4 - c^4 = 0$$

$$x, c > 0 \Rightarrow x = c, \quad c = -\lambda c^4 \Rightarrow \lambda = -\frac{1}{c^3}.$$

$$\text{驻点 } (x, y, z, t, \lambda) = (c, c, c, c, -\frac{1}{c^3}).$$

$$L = x + y + z + t + \lambda(xyzt - c^4)$$

$$dL = dx + dy + dz + dt$$

$$+ \lambda(yzt dx + xzt dy + xyt dz + xyz dt)$$

$$\begin{aligned} d^2 L = & -\frac{1}{c^3} [(ztdy + ytdz + yzdt) dx \\ & + (ztdx + xtdz + xzdt) dy \\ & + (ytdx + xtdy + xydt) dz \\ & + (yzt dx + xzt dy + xydz) dt] \end{aligned}$$

在 $x=y=z=t=c$ 处:

$$d^2L = -\frac{2}{c} [dx dy + dy dz + dz dx + \overset{t=x=y=z}{dt} (dx+dy+dz)]$$

在 $xyz t = c^4$ 上: $d(xyz t) = 0 \Rightarrow$

$$yzt dx + xzt dy + xyt dz + xyz dt = 0 \rightarrow$$

在 $x=y=z=t=c$ 处:

$$dx + dy + dz + dt = 0 \Rightarrow dt = -(dx + dy + dz)$$

$$d^2L \Big|_{(c,c,c,-\frac{1}{c})} = -\frac{2}{c} [dx dy + dy dz + dz dx - (dx + dy + dz)^2]$$

$$= \frac{1}{c} \left((dx + dy + dz)^2 + dx^2 + dy^2 + dz^2 \right) > 0$$

$$\forall (dx, dy, dz) \neq \vec{0} \text{ 时: } \rho = \sqrt{dx^2 + dy^2 + dz^2}$$

$$(x, y, z, t) = (c, c, c, c) + (dx, dy, dz, dt)$$

$$\underline{f(x, y, z, t) - f(c, c, c, c) = \frac{1}{2} d^2L \Big|_{(c,c,c,-\frac{1}{c})} + o(\rho^2)} > 0$$

$$\Rightarrow f(c, c, c, c) = 4c \text{ 为极大值. } \#$$

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